

◆ INTERMEDIATE PYTHON PRACTICE (MNC LEVEL)

1 String & Logic (Very Important)

1. Reverse a string **without using slicing or reverse()**.
 2. Check if two strings are **anagrams**.
 3. Find the **first non-repeating character** in a string.
 4. Count frequency of each character **without using Counter**.
 5. Remove duplicate characters but **keep order**.
 6. Find the **longest word** in a sentence.
 7. Check whether a string is **rotated** version of another.
 8. Compress string: "aaabbc" → "a3b2c1".
 9. Find all **palindromic substrings**.
 10. Convert "i am learning python" → "I Am Learning Python" **without using title()**.
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2 List / Array Logic (MNC Favorite)

11. Find the **second largest number** without sorting.
12. Move all **zeros to end** (order preserved).
13. Find **missing number** in a range 1–n.
14. Find **duplicate elements** without extra list.

15. Rotate list by **k** positions.
 16. Find **pair whose sum is equal to target**.
 17. Find **subarray with maximum sum**.
 18. Remove duplicates from list **without set()**.
 19. Merge two sorted lists **without sort()**.
 20. Find **intersection** of two lists.
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3 Dictionary & Data Handling

21. Count word frequency in a paragraph.
 22. Find student with **highest average marks**.
 23. Invert dictionary (keys → values, values → keys).
 24. Group words by length.
 25. Find key with **maximum value**.
 26. Sort dictionary by values.
 27. Merge two dictionaries **without update()**.
 28. Count how many values are repeated.
 29. Create a dictionary from two lists.
 30. Remove keys with value = 0.
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4 Function & Functional Programming

31. Write function that returns **prime numbers** from list.
 32. Implement `map()` using loop.
 33. Implement `filter()` using loop.
 34. Implement `reduce()` using loop.
 35. Write function that accepts **variable arguments**.
 36. Create decorator to **measure execution time**.
 37. Create decorator to **check login**.
 38. Write recursive function for **factorial & fibonacci**.
 39. Validate password (upper, lower, digit, special char).
 40. Create custom exception for invalid age.
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◆ **ADVANCED PYTHON PRACTICE (REAL COMPANY LOGIC)**

5 Problem Solving (Product Based Companies)

41. Find **longest consecutive sequence** in array.
42. Detect loop in a list.
43. Implement **binary search** without recursion.
44. Implement **LRU cache logic**.
45. Find majority element ($> n/2$).
46. Find all permutations of a string.

- 47. Solve **two sum** without using extra space.
 - 48. Check if array can be divided into equal sum subsets.
 - 49. Implement **stack using list**.
 - 50. Implement **queue using stack**.
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6 Real-World Scenario Questions (MNC Style)

- 51. ATM system (withdraw, balance, insufficient funds).
 - 52. Login system with **3 attempts only**.
 - 53. Employee management system (add, update, delete).
 - 54. Online shopping cart logic.
 - 55. Cab fare calculation based on distance & time.
 - 56. Bank interest calculation system.
 - 57. Student result processing system.
 - 58. Library management system.
 - 59. Online exam system.
 - 60. Movie ticket booking system.
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7 File Handling & Data Processing

- 61. Count words in a file.
- 62. Find longest line in a file.

- 63. Remove duplicate lines from file.
 - 64. Read CSV and calculate average.
 - 65. Log file error analysis.
 - 66. Merge two files.
 - 67. Search keyword in file.
 - 68. Replace word in file.
 - 69. Create backup of file.
 - 70. Sort file content.
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SQL + Python (MNC Combo)

- 71. Read data from database and analyze.
 - 72. Convert SQL result to dictionary.
 - 73. Find duplicate records.
 - 74. Pagination logic.
 - 75. Data validation before insert.
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PYTHON FULL MIX COMBINATION QUESTIONS

(List + String + Dict + Function + Loop + Logic + Real-World)

◆ LEVEL 1: INTERMEDIATE MIX (INTERVIEW BASE)

1. Ek sentence diya hai.
 - Duplicate words hatao
 - Order same rakho
 - Word count dictionary banao
 2. Students ka data list of dictionaries me hai.
 - Har student ka average marks nikalo
 - Topper find karo
 - Fail students alag list me rakho
 3. Ek list of numbers di hai.
 - Even numbers filter karo
 - Unka square nikalo
 - Final result dictionary me store karo (number → square)
 4. Ek paragraph diya hai.
 - Word frequency nikalo
 - Sabse zyada aane wala word find karo
 - Count ke hisab se sort karo
 5. Ek list me integers + strings mix hain.
 - Sirf integers lo
 - Positive numbers filter karo
 - Second largest number find karo
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◆ LEVEL 2: INTERMEDIATE + LOGIC MIX

6. Ek string di hai.

- Har word reverse karo
- Sentence ka structure same rakho
- Output list me return karo

7. Products ka data dictionary me hai.

- Out of stock items find karo
- Price pe 18% GST add karo
- Total inventory value nikalo

8. Employees ka data list of dictionaries me hai.

- Salary > 50,000 wale filter karo
- Department wise group karo
- Har department ka average salary nikalo

9. Ek list di hai.

- Duplicate remove karo (order preserved)
- Palindrome numbers alag list me rakho
- Result dictionary return karo

10. Ek sentence diya hai.

- Longest word find karo
 - Check karo palindrome hai ya nahi
 - Word length ke hisab se dictionary banao
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◆ **LEVEL 3: ADVANCED MIX (PRODUCT COMPANY STYLE)**

11. Ek list of numbers di hai.

- Consecutive sequence find karo
- Uska length return karo
- Sequence list me do

12. Transaction records list of dictionaries me hain.

- User-wise total amount calculate karo
- Highest spender find karo
- Sorted summary banao

13. Ek string di hai.

- All permutations generate karo
- Palindromic permutations alag rakho
- Count return karo

14. Ek list di hai.

- Subarrays nikalo
- Maximum sum wala subarray find karo
- Sum aur subarray dono return karo

15. Login attempts list di hai.

- 3 se zyada fail attempts wale users block karo
- Success rate calculate karo
- Final report dictionary me return karo

◆ LEVEL 4: REAL-WORLD + SYSTEM THINKING MIX



16. ATM system banao using functions.

- Balance check
- Withdraw logic
- Invalid amount handle
- Transaction history maintain karo (list + dict)

17. Shopping cart system.

- Add / remove products
- GST + discount calculate
- Final bill dictionary me return karo

18. Online exam system.

- Questions list
- Answers check
- Score calculate
- Result dictionary generate karo

19. Employee attendance system.

- Punch in / out times list me store
- Working hours calculate
- Monthly summary dictionary banao

20. Bank account system.

- Deposit / withdraw
 - Interest calculation
 - Transaction log maintain karo
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◆ **LEVEL 5: INTERVIEW BRAIN-TESTER (🔥 HARD)**

21. Ek sentence diya hai.

- Har word ka character frequency nikalo
- Most complex word find karo (max unique chars)

22. Ek list of numbers di hai.

- Rotate list
- Rotation ke baad pairs find karo jinka sum target ho

23. Employee data diya hai.

- Duplicate IDs remove karo
- Salary normalize karo
- Department-wise ranking banao

24. Log file entries list me hain.

- Error count nikalo
- Most frequent error find karo
- Report dictionary banao

25. Mixed data list di hai.

- Numbers → sum

- Strings → concatenate
 - Booleans → count
 - Final result dictionary me do
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PRACTICE RULE (VERY IMPORTANT)

-  Direct solution mat likho
-  Pehle **steps** likho
-  Phir code
-  Edge cases handle karo
-  Time complexity socho